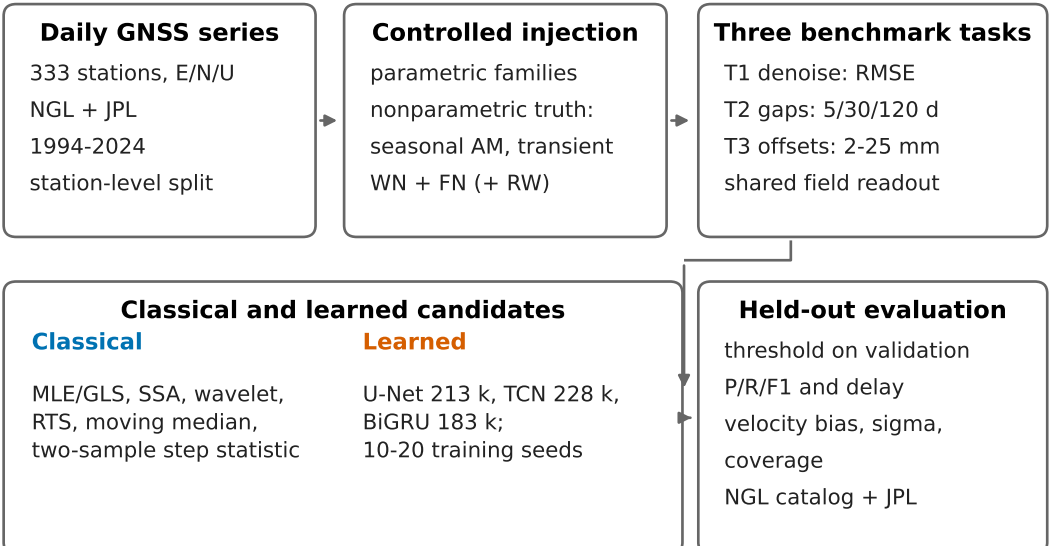
Daily GNSS series
333 stations, E/N/U
NGL + JPL
1994-2024
station-level split
Controlled injection
parametric families
nonparametric truth:
seasonal AM, transient
WN + FN (+ RW)
Three benchmark tasks
T1 denoise: RMSE
T2 gaps: 5/30/120 d
T3 offsets: 2-25 mm
shared field readout
Classical and learned candidates
Classical
MLE/GLS, SSA, wavelet,
RTS, moving median,
two-sample step statistic
Learned
U-Net 213 k, TCN 228 k,
BiGRU 183 k;
10-20 training seeds
Held-out evaluation
threshold on validation
P/R/F1 and delay
velocity bias, sigma,
coverage
NGL catalog + JPL